ASSIGNMENT 3

Q5 - Hoisting Lab: The Sequence Trap

JavaScript solution

```
// q5.js — corrected and predictable version
"use strict";

var score = 50;
function announce() {
  console.log("Game started");
}
let status = "ready";

console.log(score);
announce();

function startGame() {
  console.log(status);
}
startGame();

// Arrow-function comparison:
const announceArrow = () => console.log("Game started (arrow)");
const startGameArrow = () => console.log(status);

announceArrow();
startGameArrow();
```

Explanation / Debug notes

In the original, var score is hoisted and initialized to undefined, so the first log prints undefined. Function declarations are hoisted with their function body, so announce() works before its declaration. let status is in the temporal dead zone until its declaration executes; startGame() runs after status is initialized, so it prints ready. Arrow functions assigned to const are not usable before their initialization. The corrected version initializes variables before logging.

Run the code in the browser console or Node.js. Save each solution in its own .js file as required.